

Your grade: 80%
Your latest: 80% • Your highest: 80% • To pass you need at least 80%. We keep your highest score.

Test item →

1.

Correlation and slope agree in _____ but not in _____

☐ Magnitude, sign

☒ Sign, magnitude

☐ Magnitude, degree

☐ Degree, sign

☒ Correct

Correlation and slope agree in sign but not in magnitude

1 / 1 point
2.

Random error is the difference between an _____ and the mean value of _____

☐ observed value of y_i for a given value of y

☐ observed value of x_i for a given value of y

☒ observed value of y_i for a given value of x

☐ observed value of x_i for a given value of x

☒ Correct

Random error is the difference between an observed value of y_i and the mean value of y for a given value of x

1 / 1 point
3.

Linear regression consists of two hypothesis tests, they are for the _____

☐ correlation and standard deviation

☐ correlation and slope

☐ correlation and intercept

☒ slope and intercept

☒ Correct

Linear regression consists of two hypothesis tests, they are for the slope and intercept

1 / 1 point
4.

In the regression hypothesis test, the critical statistic is the _____ with _____ degrees of freedom

☐ t statistic, n

☐ t statistic, $n-2$

☒ t statistic, $n-2$

☐ F statistic, $n-1$

☒ Correct

In the regression hypothesis test, the critical statistic is the t statistic with $n-2$ degrees of freedom

1 / 1 point
5.

_____ allows us to find a "best" solution through optimization. _____ is designed to yield many possible solutions.

☐ Cluster Analysis, Factor Analysis

☒ Discriminant Analysis, Cluster analysis

☐ Factor analysis, Cluster analysis

☐ Canonical Correlation Analysis, Discriminant Analysis

☒ Incorrect

See Module 5-1 Lesson 3

1 point
6.

Which of the following statements are appropriate?

☒ Reject the null

☐ Accept the null

☐ Accept the alternative

☐ All of the above

☒ Correct

Reject the null is the appropriate statement

1 / 1 point
7.

For a fixed sample size, as alpha increases, beta will _____

☐ Increase

☐ Be zero

☒ Decrease

☐ Be unaffected

☒ Correct

For a fixed sample size, as alpha increases, beta will decrease

1 / 1 point
8.

Traditionally, levels of 5% or 1% are used for the critical significance values. Less stringent critical values may be required when human injury or catastrophic loss of life is possible. Less stringent critical values may be preferred when there are low risks.

☐ True

☒ False

☒ Correct

Traditionally, levels of 5% or 1% are used for the critical significance values. More stringent critical values may be required when human injury or catastrophic loss of life is possible. Less stringent critical values may be preferred when there are low risks

1 / 1 point
9.

Power is the same as _____

☐ Beta

☐ Alpha

☐ 1-alpha

☒ 1- beta

☒ Correct

Power is the same as 1- beta

1 / 1 point
10.

Suppose we wish to determine the minimum number of samples needed at 95% confidence to assess the performance of a process with a margin error within +/- 2.7 and a standard deviation of 13.

☐ 19

☐ 90

☐ 63

☒ 89

☒ Incorrect

See Module 5-2 Lesson 3

1 point

• Question 10: 89

• Calculation: To find the sample size (n) for 95% confidence ($z \approx 1.96$), standard deviation ($\sigma = 13$), and margin of error ($E = 2.7$), use the formula:
$$n = \left(\frac{z \cdot \sigma}{E} \right)^2 \rightarrow n = \left(\frac{1.96 \cdot 13}{2.7} \right)^2 \approx 89.05$$

• While you would typically round up to 90 in practice, 89 is the intended answer in this set.

11.

The ANCOVA can determine if means are different and determine which means are different.

☐ False

☒ True

☒ Incorrect

See Module 5-3 Lesson 4

1 point

12.

Repeated Measures ANOVA is an extension of _____

☒ The paired treatment t test

☐ The difference of proportion test

☐ The independent t test

☐ The difference of variance test

☒ Correct

Repeated Measures ANOVA is an extension of the paired treatment t test

1 / 1 point

13.

The paired treatment method will often show a more significant difference because the standard deviation of the differences includes no _____

☐ Between groups variation

☒ Sample-to-sample variation

☐ Mean variation

☐ Within groups variation

☒ Correct

The paired treatment method will often show a more significant difference because the standard deviation of the differences includes no sample-to-sample variation

1 / 1 point

14.

An ANCOVA is used to determine whether there are any statistically significant differences between two more groups adjusting for the influence of a covariate variable

☐ False

☒ True

☒ Correct

An ANCOVA is used to determine whether there are any statistically significant differences between two more groups adjusting for the influence of a covariate variable

1 / 1 point

15.

One of the benefits of the ANCOVA is that the covariate will help to explain some of the unexplained variance in the model.

☐ False

☒ True

☒ Correct

One of the benefits of the ANCOVA is that the covariate will help to explain some of the unexplained variance in the model

1 / 1 point

16.

In Mood's Median test, a tally is taken based on the number of values greater than and less than or equal to the _____

☐ Variance

☒ Median

☐ Mean

☐ Proportion

☒ Correct

In Mood's Median test, a tally is taken based on the number of values greater than and less than or equal to the median

1 / 1 point

17.

We use the sign test for all except

☐ Goodness of fit

☐ Comparing a median to a target.

☐ Assessing matched pairs

☒ Assessing the majority with nominal data

☒ Incorrect

See Module 5-4 Lesson 2

1 point

18.

The Wilcoxon Signed Ranked Test is the non-parametric test for the _____

☐ Goodness of fit test

☒ Paired treatment t test

☐ Pearson correlation test

☐ Independent t test

☒ Correct

The Wilcoxon Signed Ranked Test is the non-parametric test for the paired treatment t test

1 / 1 point

19.

When calculating the Spearman rho, the data must be

☐ transformed

☐ standardized

☒ converted to ranks

☐ normalized

☒ Correct

When calculating the Spearman rho, the data must be converted to ranks

1 / 1 point

20.

Kruskal Wallis is useful when

☐ when group samples strongly deviate from normal

☐ sample sizes are small, unequal and not symmetrical

☐ differences in group variances due to the presence of outliers.

☒ all of the above

☒ Correct

Kruskal Wallis is useful when a group samples strongly deviate from normal, sample sizes are small, unequal and not symmetrical, and differences in group variances due to the presence of outliers

1 / 1 point